



الجامعة الإسلامية للتكنولوجيا
UNIVERSITE ISLAMIQUE DE TECHNOLOGIE
ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

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Course Title: Autocad

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Department: EEE

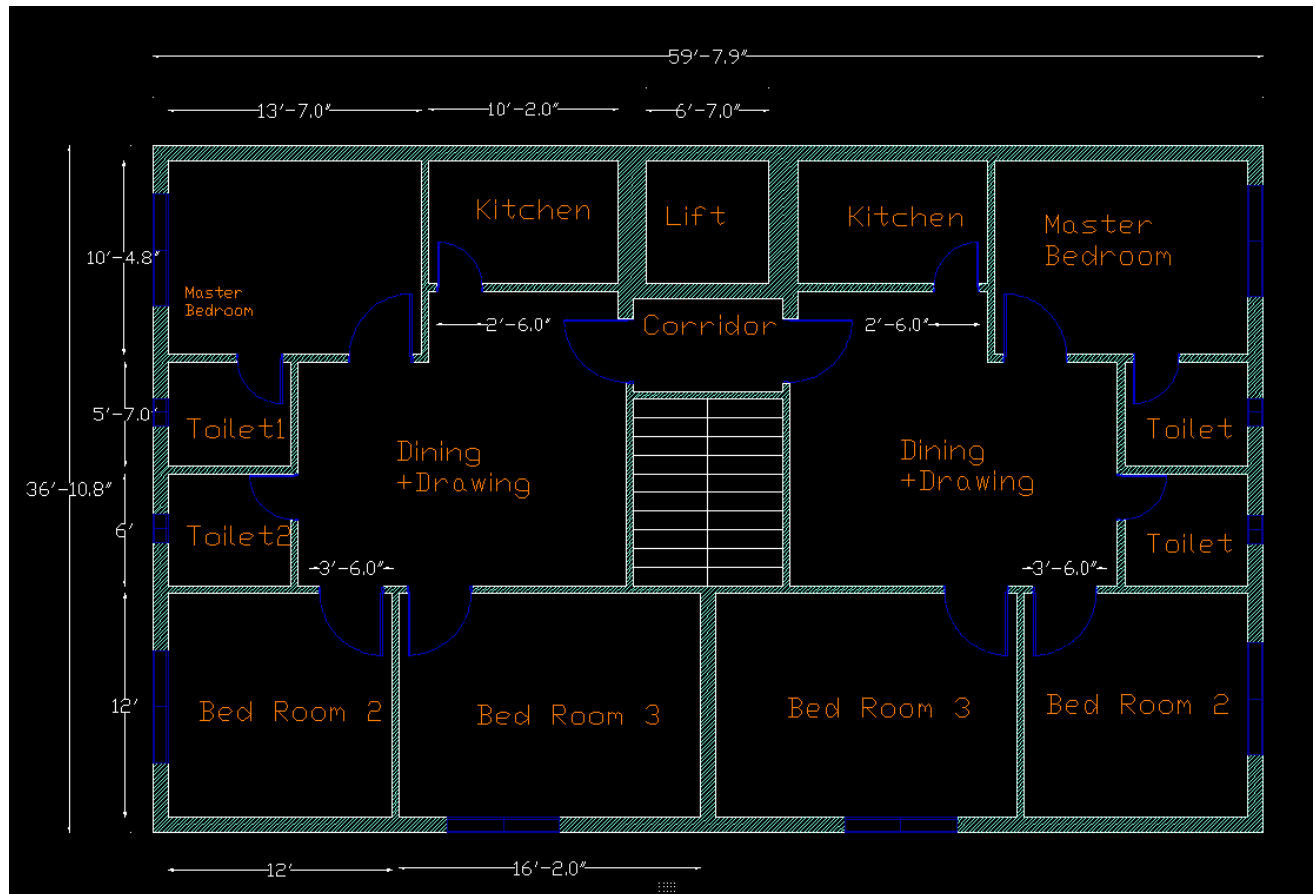
Section: A1

Group: 03

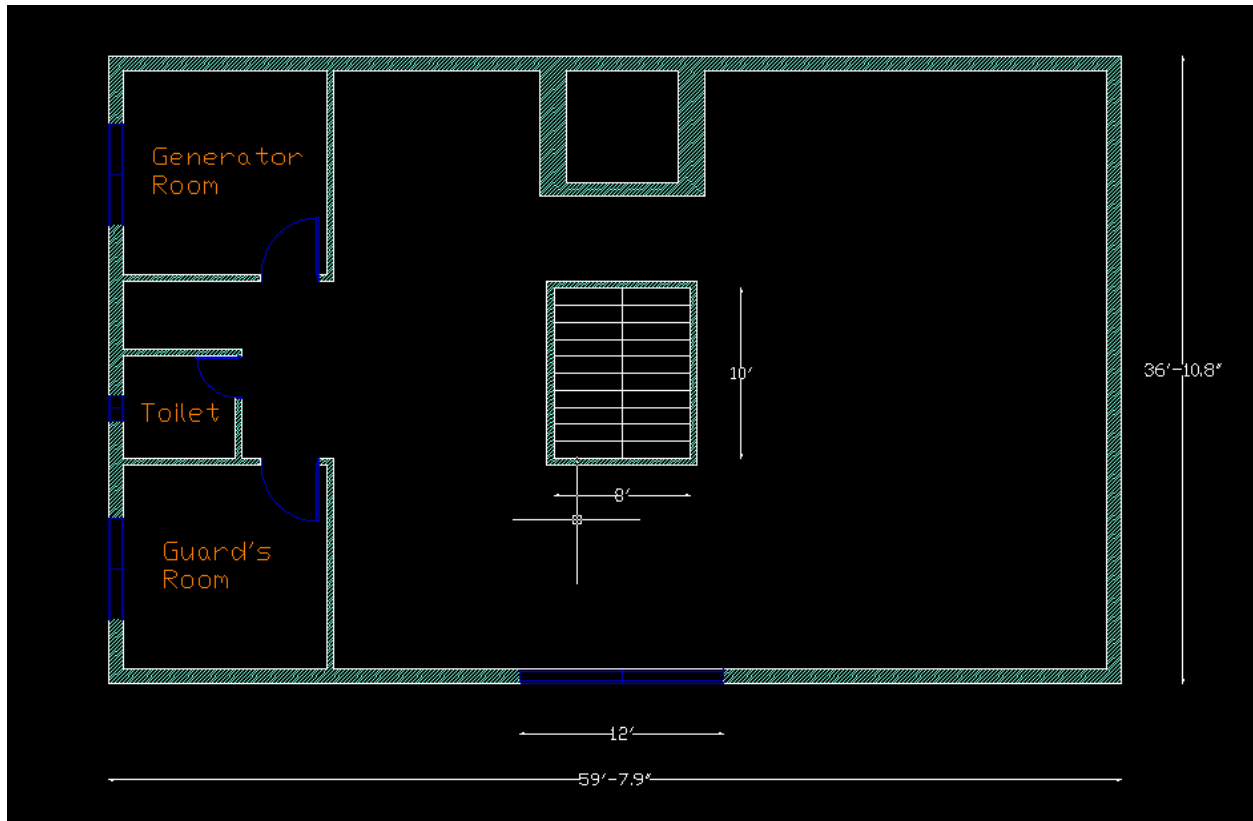
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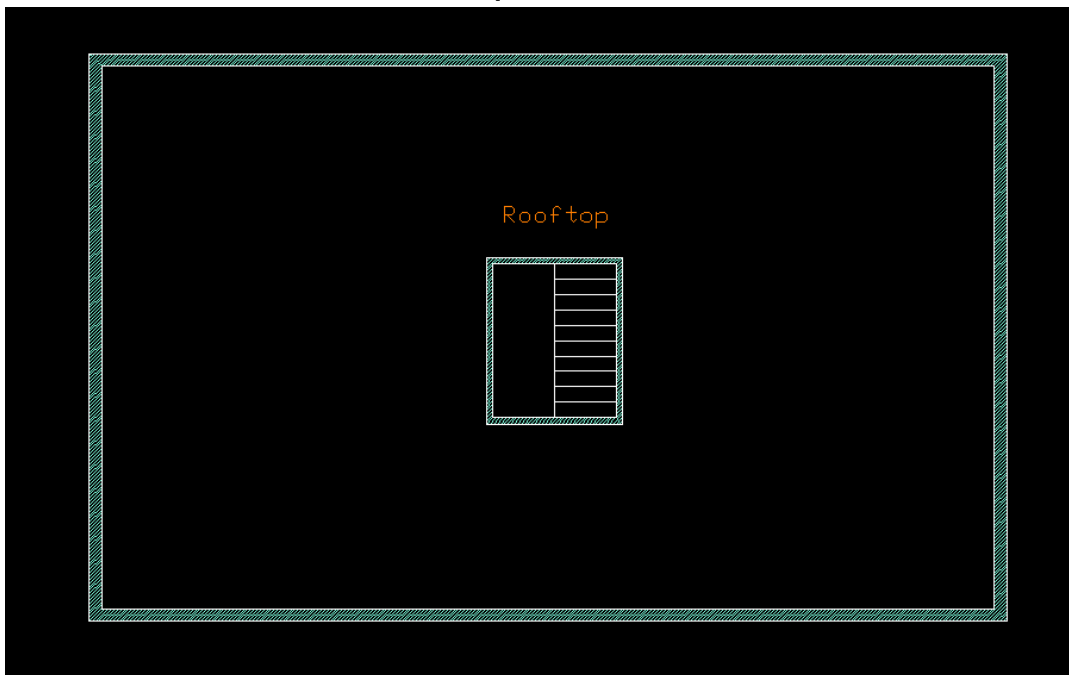
Floor Plan



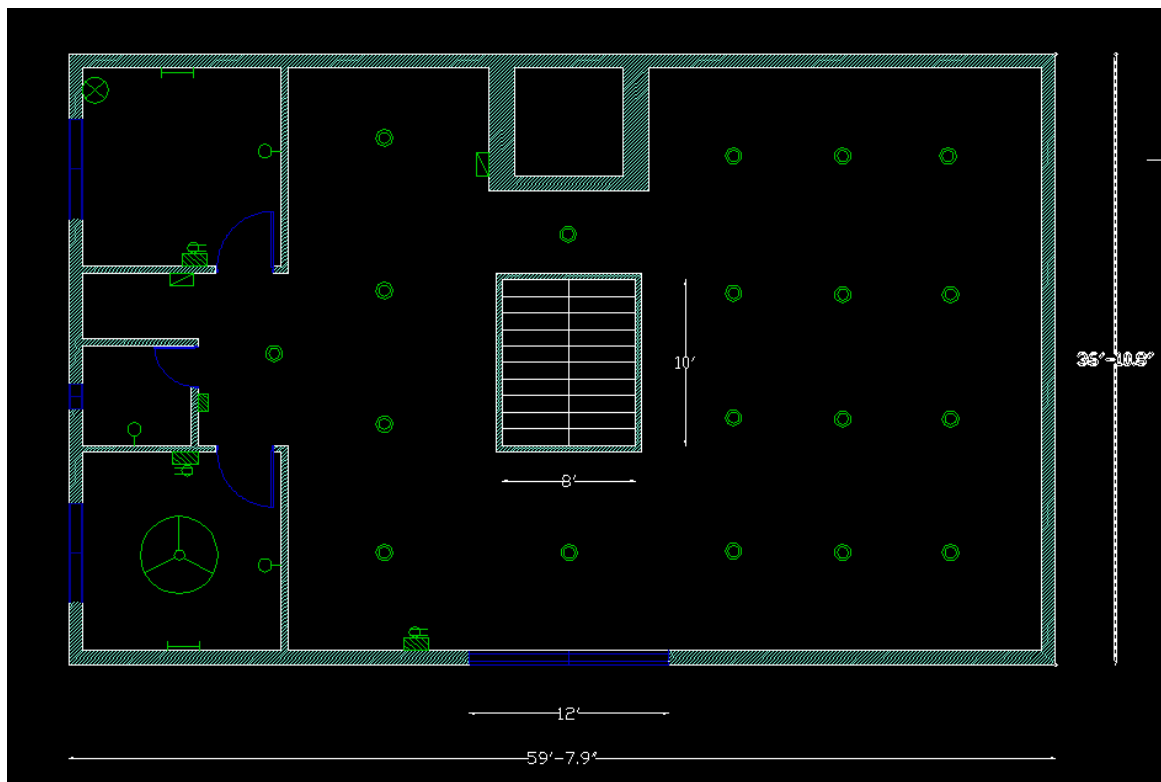
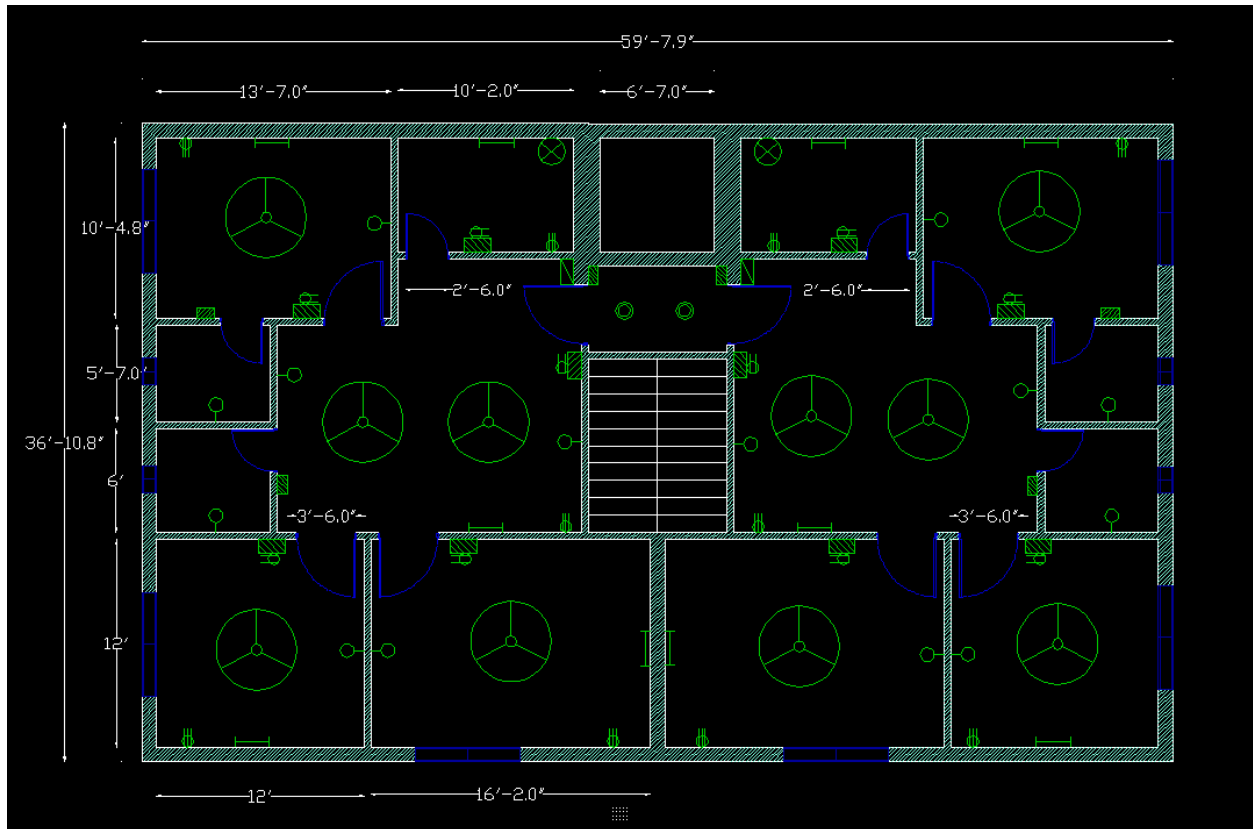
Garage



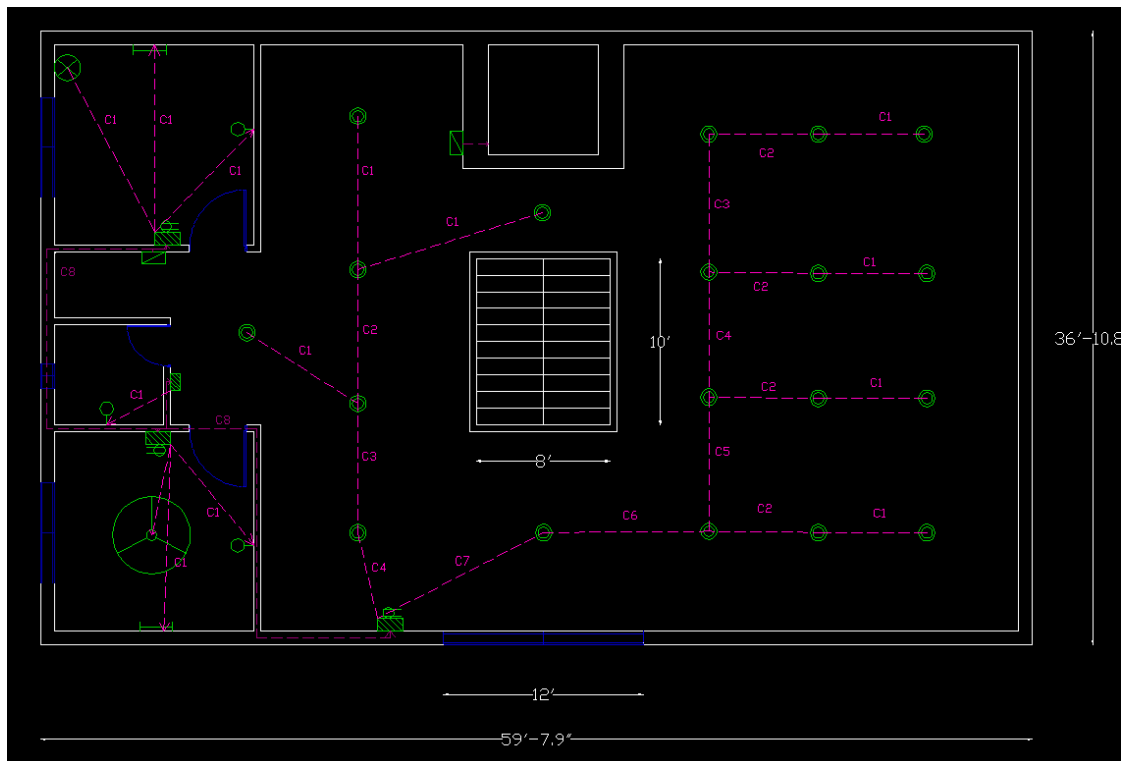
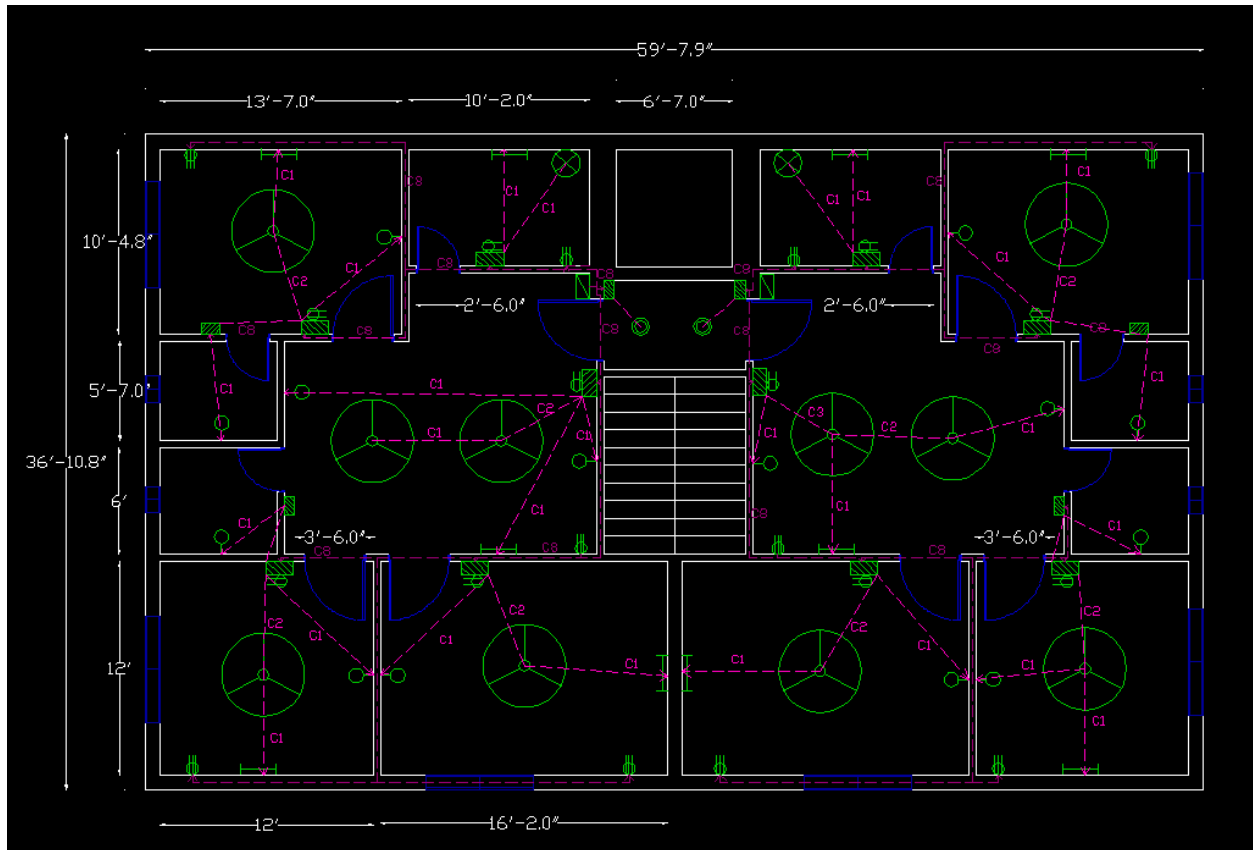
Top Floor



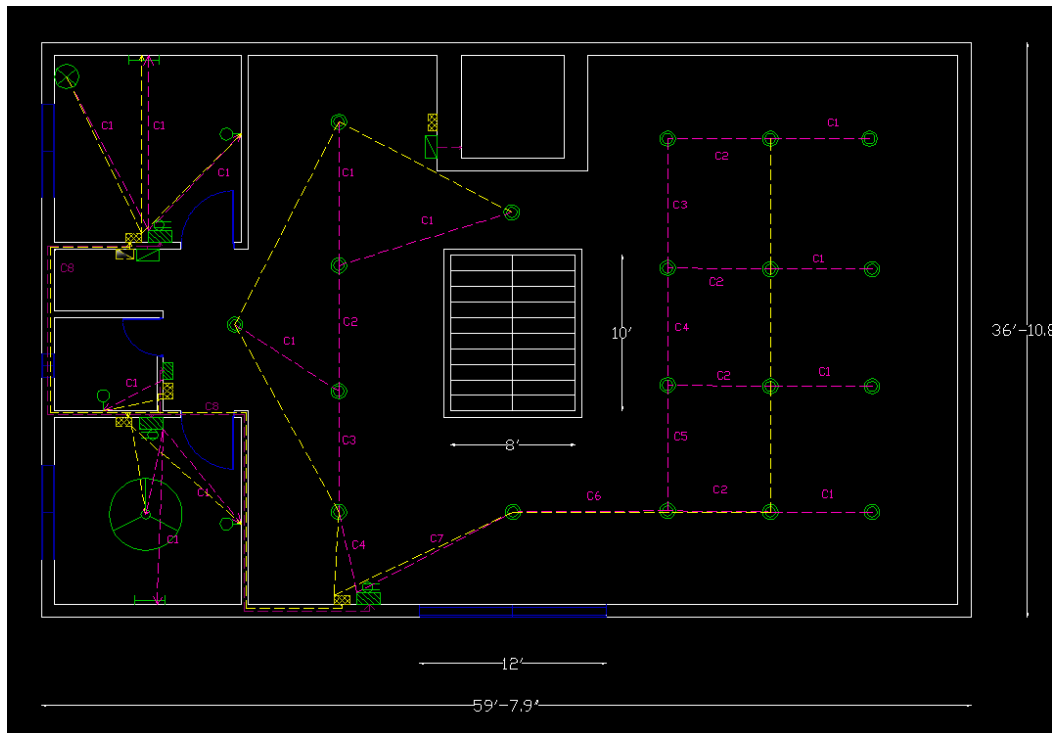
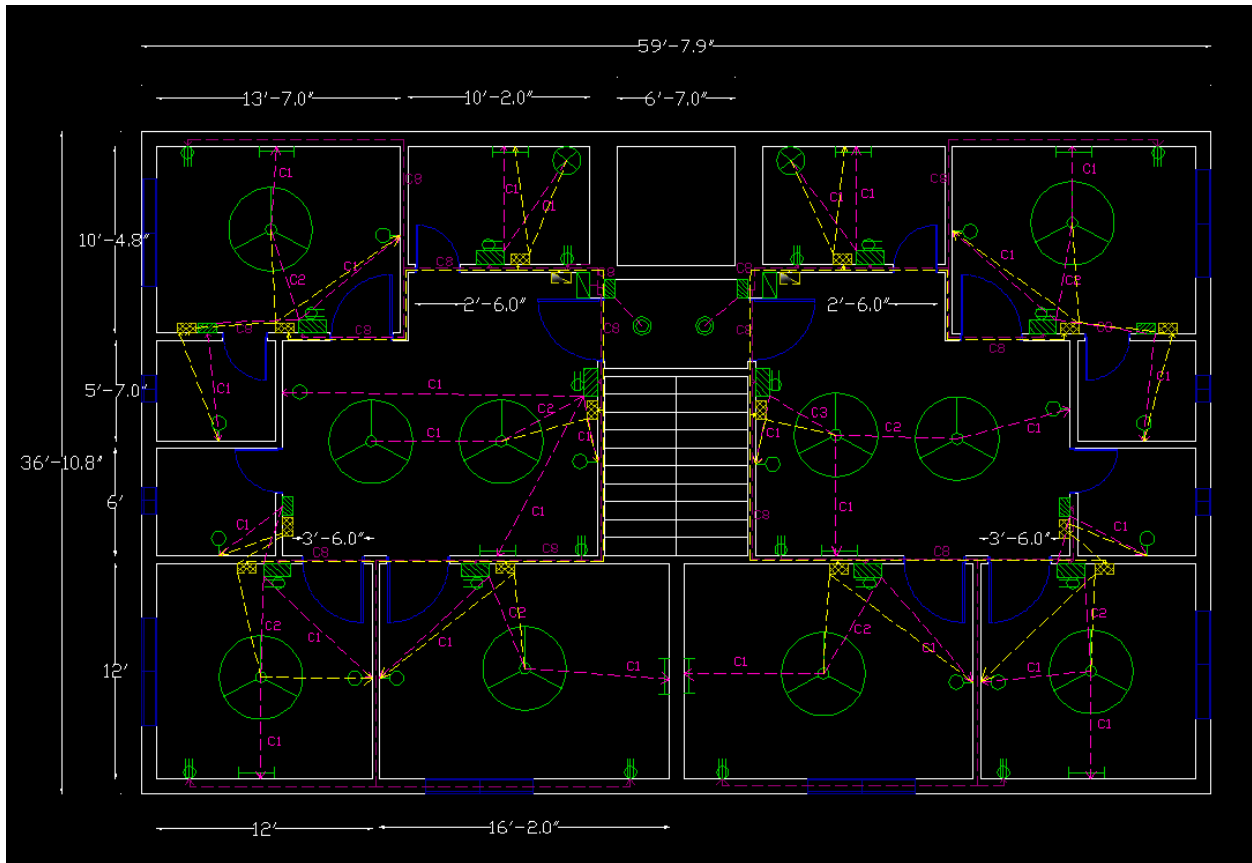
Fittings and Fixtures



Conduction Layout



Conduction Layout with Emergency line

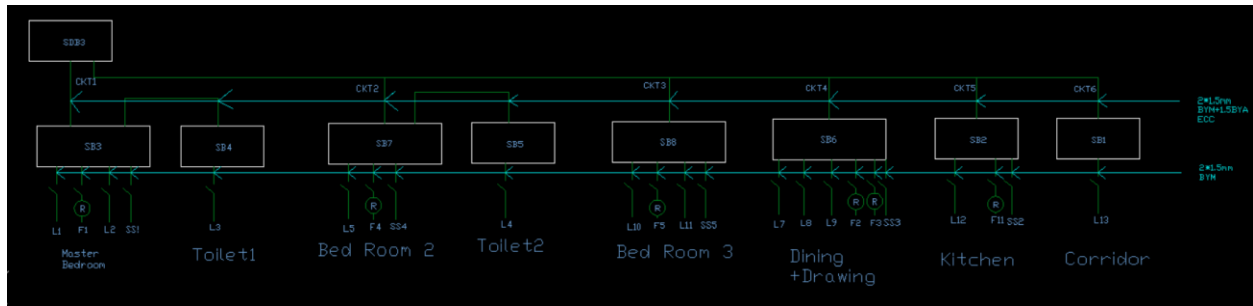


LEGENDS

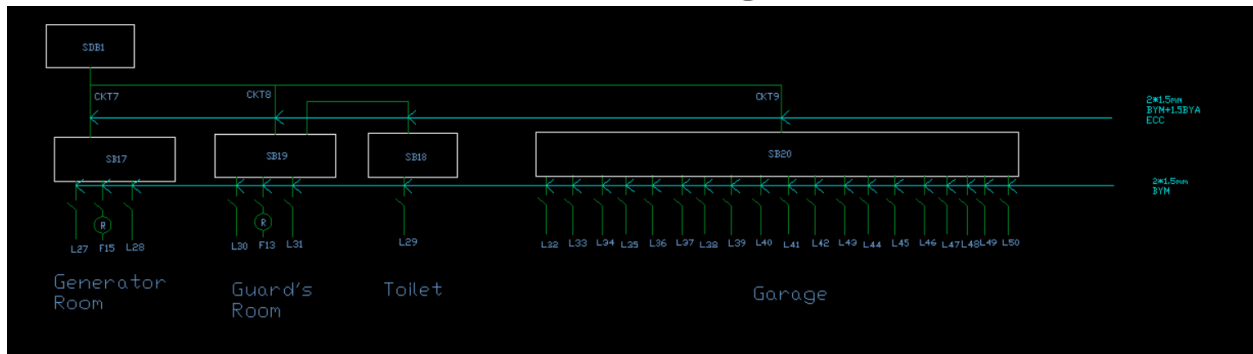
DESCRIPTION	HEIGHT	CAPTION	SYMBOL
4'-20W wall mounted flurosent tube light	Lintel	TB	
20W Fluorescent light bulb	Lintel	LB	
20W Fluorescent ceiling light	Ceiling	LC	
56" 70W sweep fan	Ceiling	F	
Switch Board	Mid Wall	SB	
Emergency Switch Board	Mid Wall	ESB	
Sub Distribution Board	Lintel	SDB	
Emergency Sub Distribution Board	Lintel	ESDB	
Exhaust Fan	Lintel	EX	
Power Socket for PC	Lower wall	P 15 A	
Power Socket for AC, Heater, etc	Lower wall	Q 20 A	

Legends for Conduits	Conduit size
C1= 2 x 1.5 rm BYM	3/4"
C 2 = 4 * 1.5rm BYM	3/4"
C3 = 6x rm BYM	3/4"
C4 = 2 * 1.5rm BYM + 1.5 rm BYA ECC	3/4"
C5 = 2 * 1.5 rm BYM + 1.5 rm BYA ECC	3/4"
C6= 2 * 4 rm BYM + 4 rm BYA ECC	3/4"
C7 = 2 * 6 rm BYM + 6 rm BYA ECC	1"
C8 = 4 * 4rm BYM rm BYA ECC	1"
C9 = 4 * 6 rm BYM + 6 rm BYA ECC	1"

Switch Board Connection Diagram (Main Floor)



Switch Board Connection Diagram (Ground Floor)



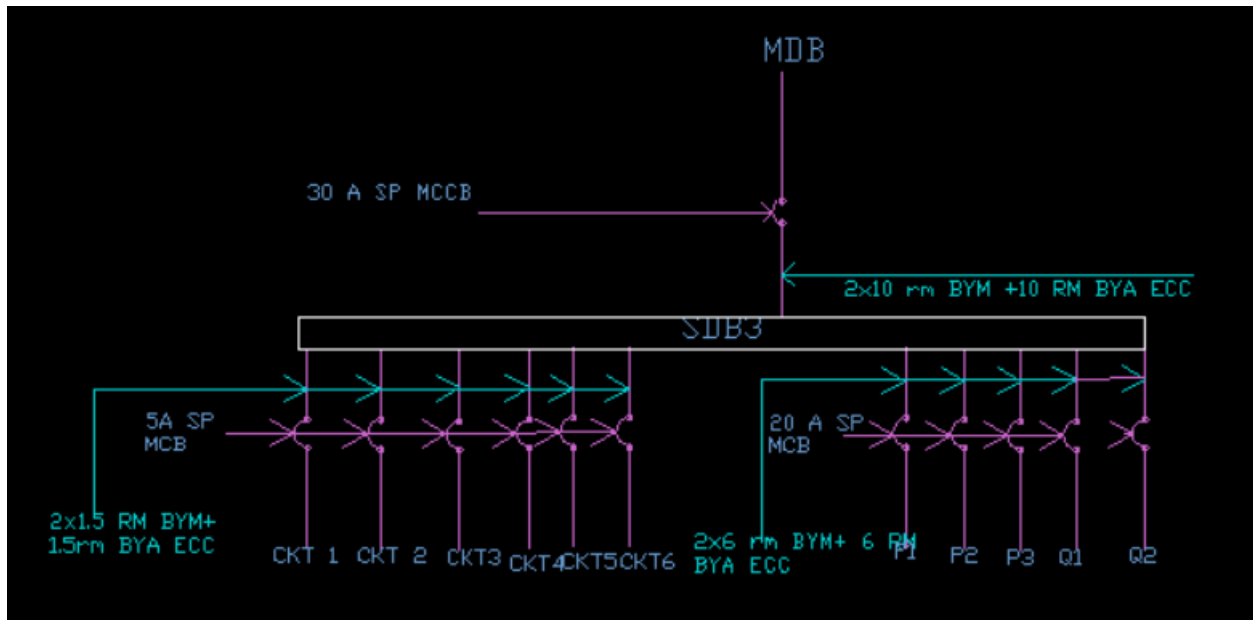
The diagram illustrates a single-line wiring system for a 240V AC supply. The main supply line is at the top, with five circuit breakers (CKT1 to CKT5) distributing power to different rooms. Each room has its own electrical panel (ESB3 to ESB8) and is connected to the main supply line via a circuit breaker. The diagram also shows various outlets (F1 to F11) and switches (L1 to L12) connected to the room panels. A legend on the right indicates that the blue line represents 2x1.5mm BYN+1.5BYA EEC and the red line represents 2x1.5mm BYM.

Legend:

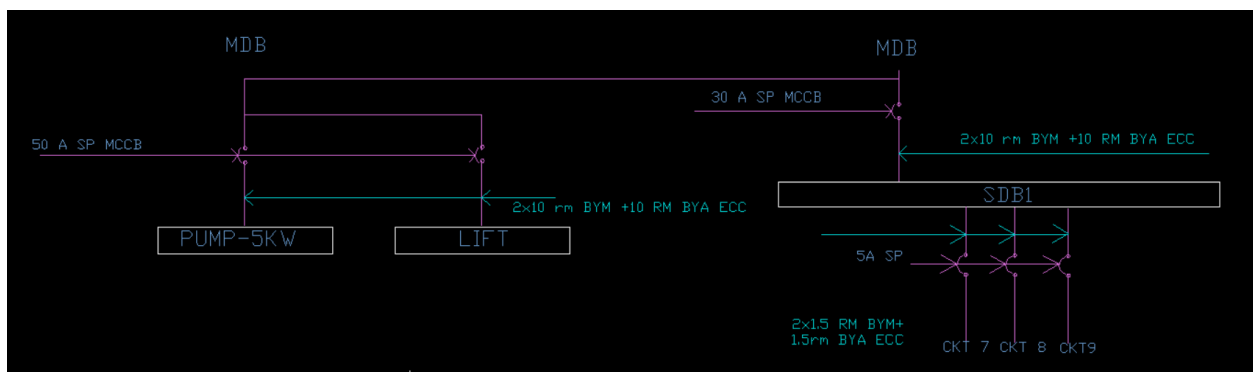
- Blue line: 2x1.5mm BYN+1.5BYA EEC
- Red line: 2x1.5mm BYM

[illegible]

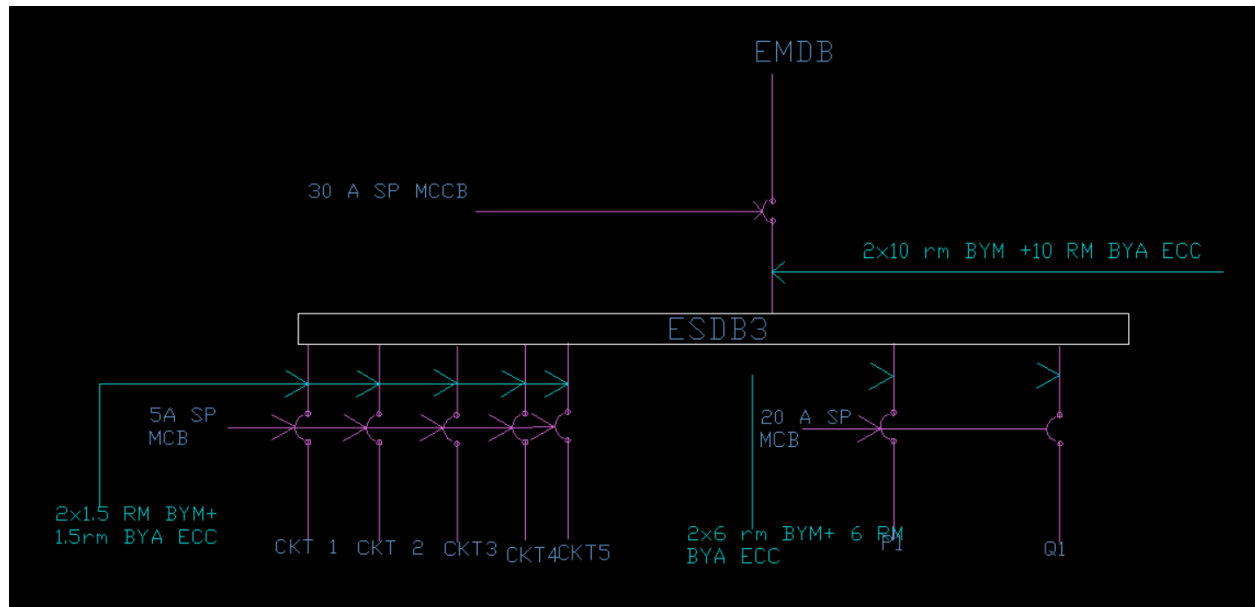
Sub Distribution Board Diagram



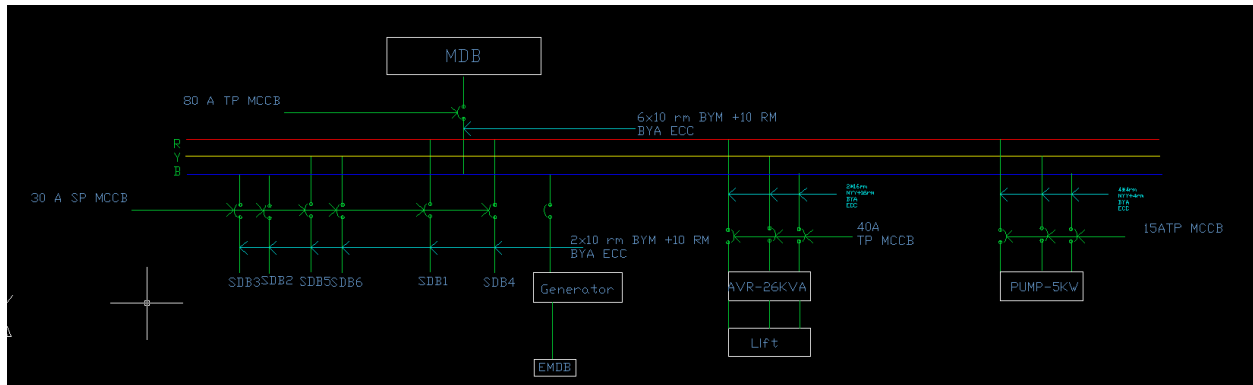
Sub Distribution Board Diagram Ground Floor



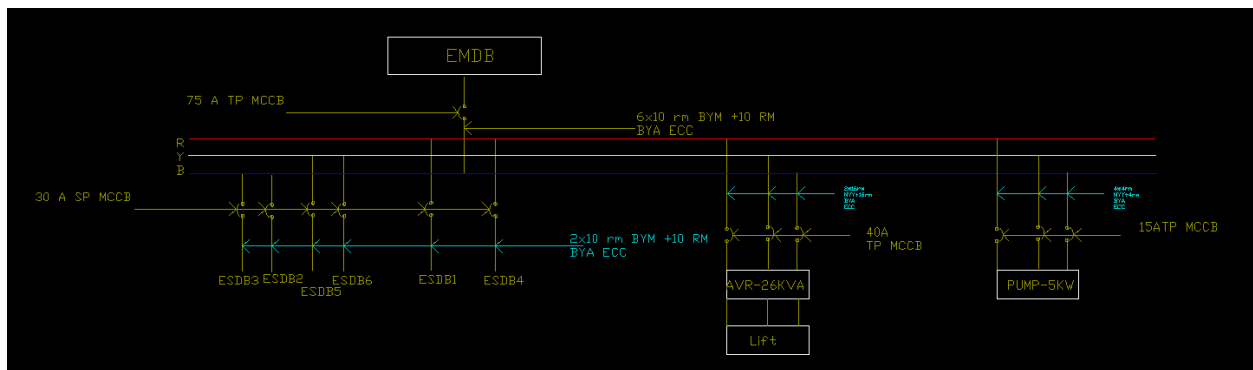
Emergency Sub Distribution Board Diagram



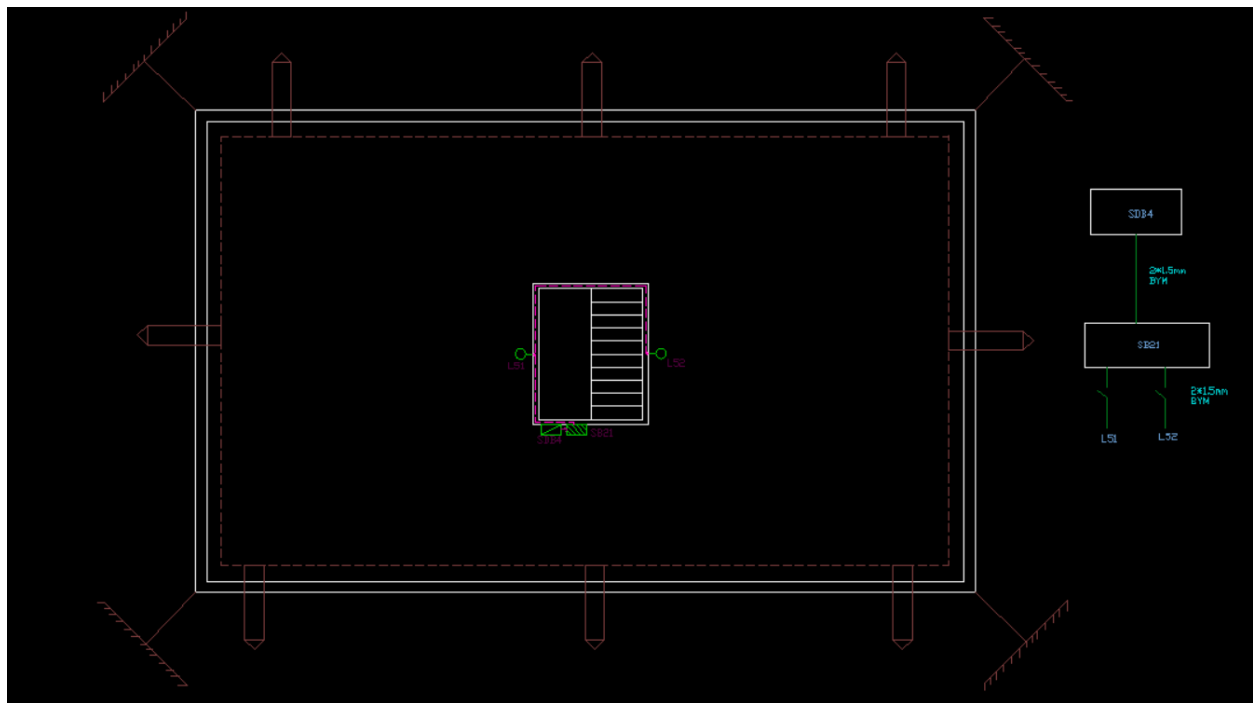
Main Distribution Board Diagram



Emergency Main Distribution Board Diagram



Lightning Protection



Calculations for Light Bulbs (LB) and Fans(F)

One 56-inch diameter fan for every 100 sqft.

Bedroom 1

A= 13 x 10 sqft.

Number of Fans:

Room area = 13 ft * 10 ft = 130 sqft

Number of fans = 130 sqft / 100 sqft/fan = 1.3

So 1 Fan

Number of Light Bulbs:

$$E = (n * N * F * UF * LLF) / A$$

where:

E = Illuminance (standard for bedrooms is 100 lumens/m²)

LLF x UF (Light Loss Factor x Utilization Factor) = 0.7 (assumed value)

n (number of lights per luminaire) = 1 (assuming single bulbs)

Flux (lumens per bulb) = 1250 lumens (example for a 20W energy-saving bulb)

A (Area in square meters) = room area from sqft to m²: 130 sqft * 0.092903 m²/sqft ≈ 12.08 m²

$$N = (E * A) / (LLF * UF * n * Flux)$$

$$N = (100 \text{ lumens/m}^2 * 12.08 \text{ m}^2) / (0.7 * 1 * 1250 \text{ lumens})$$

$$N \approx 1.38$$

So 2 bulbs

Number of fans 1

Number of lightbulbs 2

Bedroom 2

Room area = 12 ft * 12 ft = 144 sqft

Number of Fans:

Number of fans = 144 sqft / 100 sqft/fan = 1.44

So 1 fan

Number of Light Bulbs:

formula: $E = (n * N * F * UF * LLF) / A$

E (Illuminance) = 100 lumens/m² (standard for bedrooms)

LLF x UF (Light Loss Factor x Utilization Factor) = 0.7

n (number of lights per luminaire) = 1 (assuming single bulbs)

Flux (lumens per bulb) = 1250 lumens (example for a 20W energy-saving bulb)

A (Area in square meters) = room area from sqft to m²: 144 sqft * 0.092903 m²/sqft ≈ 13.38 m²

Solve for N (number of lights):

$N = (E * A) / (LLF * UF * n * Flux)$

$N = (100 \text{ lumens/m}^2 * 13.38 \text{ m}^2) / (0.7 * 1 * 1250 \text{ lumens})$

$N \approx 1.70$

So 2 bulbs

Number of fans 1

Number of lightbulbs 2

Bedroom 3

Bedroom area = 12 ft * 16 ft = 192 sqft

Number of Fans:

Number of fans = 192 sqft / 100 sqft/fan = 1.92

So 1 fan

Number of Light Bulbs:

We'll use the given formula: $E = (n * N * F * UF * LLF) / A$

where:

E (Illuminance) = 100 lumens/m² (standard for bedrooms)

LLF x UF (Light Loss Factor x Utilization Factor) = 0.7

n (number of lights per luminaire) = 1 (assuming single bulbs)

Flux (lumens per bulb) = 1250 lumens (example for a 20W energy-saving bulb)

A (Area in square meters) = room area from sqft to m²: 192 sqft * 0.092903 m²/sqft ≈ 17.84 m²

Solve for N (number of lights):

$N = (E * A) / (LLF * UF * n * Flux)$

$N = (100 \text{ lumens/m}^2 * 17.84 \text{ m}^2) / (0.7 * 1 * 1250 \text{ lumens})$

$N \approx 2.07$

So 2 bulbs

Number of fans 1

Number of lightbulbs 2

Dining+ Drawing

Room area = 12 ft * 14 ft = 168 sqft

Number of Fans:

Number of fans = 168 sqft / 100 sqft/fan = 1.68

So 2 fans

Number of Light Bulbs:

$$E = (n * N * F * UF * LLF) / A$$

E (Illuminance) = 100 lumens/m² (standard for bedrooms)

LLF x UF (Light Loss Factor x Utilization Factor) = 0.7

n (number of lights per luminaire) = 1 (assuming single bulbs)

Flux (lumens per bulb) = 1250 lumens (example for a 20W energy-saving bulb)

A (Area in square meters) = room area from sqft to m²: 168 sqft * 0.092903 m²/sqft ≈ 15.61 m²

Solve for N (number of lights):

$$N = (E * A) / (LLF * UF * n * Flux)$$

$$N = (100 \text{ lumens/m}^2 * 15.61 \text{ m}^2) / (0.7 * 1 * 1250 \text{ lumens})$$

$$N \approx 2.07$$

So 2 bulbs

We use 1 extra Tube light as Dining & Drawing room generally need more light.

Number of fans 2

Number of lightbulbs 3

Toilet 1

Toilet area = 5.7 ft * 6.7 ft = 38.19 sqft

Convert sqft to m²:

Toilet area in m² = 38.19 sqft * 0.092903 m²/sqft ≈ 3.55 m²

Light Bulb Calculation:

$$E = (n * N * F * UF * LLF) / A$$

E (Illuminance) = 100 lumens/m² (standard for toilets)

LLF x UF (Light Loss Factor x Utilization Factor) = 0.7

n (number of lights per luminaire) = 1 (assuming single bulb)

Flux (lumens per bulb) = 1250 lumens (example for a 20W energy-saving bulb)

A (Area in square meters) = 3.55 m²

Solve for N (number of lights):

$$N = (E * A) / (LLF * UF * n * Flux)$$

N = (100 lumens/m² * 3.55 m²) / (0.7 * 1 * 1250 lumens)

N ≈ 0.42

Number of lightbulbs 1

Toilet 2

Toilet area = 6 ft * 6 ft = 36 sqft

Convert sqft to m²:

Toilet area in m² = 36 sqft * 0.092903 m²/sqft ≈ 3.34 m²

Light Bulb Calculation:

$$E = (n * N * F * UF * LLF) / A$$

E (Illuminance) = 100 lumens/m² (standard for toilets)

LLF x UF (Light Loss Factor x Utilization Factor) = 0.7

n (number of lights per luminaire) = 1 (assuming single bulb)

Flux (lumens per bulb) = 1250 lumens (example for a 20W energy-saving bulb)

A (Area in square meters) = 3.34 m² (calculated previously)

Solve for N (number of lights):

$$N = (E * A) / (LLF * UF * n * Flux)$$

$$N = (100 \text{ lumens/m}^2 * 3.34 \text{ m}^2) / (0.7 * 1 * 1250 \text{ lumens})$$

$$N \approx 0.40$$

Number of lightbulbs 1

Kitchen

Toilet area = 10 ft * 6.7 ft = 67 sqft

Convert sqft to m²:

Toilet area in m² = 67 sqft * 0.092903 m²/sqft ≈ 6.22 m²

Light Bulb Calculation:

$$E = (n * N * F * UF * LLF) / A$$

where:

E (Illuminance) = 200 lumens/m² (standard for kitchen)

LLF x UF (Light Loss Factor x Utilization Factor) = 0.7

n (number of lights per luminaire) = 1 (assuming single bulb)

Flux (lumens per bulb) = 1250 lumens (example for a 20W energy-saving bulb)

A (Area in square meters) = 6.22 m² (calculated previously)

Solve for N (number of lights):

$$N = (E * A) / (LLF * UF * n * Flux)$$

$$N = (200 \text{ lumens/m}^2 * 6.22 \text{ m}^2) / (0.7 * 1 * 1250 \text{ lumens})$$

$$N \approx 1.60$$

Number of lightbulbs 1

Calculations for conduits:

Formula for ampere rating, $I = P/(V \cdot pf)$

$Pf = 0.7$

Energy saving bulb = 20W

Tubelight = 20W

Ceiling fan = 100W

Ceiling light = 20W

Switchboard socket = 100W

Lift = 6000W

Pump = 5000W

To switch board (SB):

At floors,

$$I = (6 \cdot 100 + 13 \cdot 20) / (220 \cdot 0.7)$$

$$= 5.58A$$

So, 2*2.5 rm BYM + 2.5 BYA ECC are used

In the garage,

$$I = (2 \cdot 100 + 23 \cdot 20) / (220 \cdot 0.7)$$

$$= 4.285A$$

So, 2*2.5 rm BYM + 2.5 BYA ECC are used

For lift,

$$I = (1 \cdot 6000) / (220 \cdot 0.7)$$

$$= 38.96A$$

So, 6*10 rm BYM + 2.5 BYA ECC are used

For pump,

$$I = (1*5000)/(220*0.7)$$

$$= 32.46$$

So, 4*10 rm BYM + 2.5 BYA ECC are used

At rooftop,

$$I = (2*20)/(220*0.7)$$

$$= 0.26A$$

So, 2*1.5 rm BYM + 1.5 BYA ECC are used

To sub distribution board:

For floor,

$$I = (6*100 + 13*20)/(220*0.7)$$

$$= 5.58A$$

So, 2*2.5 rm BYM + 2.5 BYA ECC are used

$$\text{SDB load} = (1020*0.7 + 6*3000)$$

$$= 4314W$$

$$I = (4314/(220*0.7))$$

$$= 28 A$$

So, 30A SP MCCB is needed from SDB to MDB

In the garage,

$$I = (2*100 + 23*20)/(220*0.7)$$

$$= 4.285A$$

So, 2*2.5 rm BYM + 2.5 BYA ECC are used

For lift,

$$I = (1*6000)/(220*0.7)$$

$$= 38.96A$$

So, 6*10 rm BYM + 2.5 BYA ECC are used

For pump,

$$I = (1*5000)/(220*0.7)$$

$$= 32.46A$$

So, 4*10 rm BYM + 2.5 BYA ECC are used

$$I = 3.24 + 38.96 + 32.46$$

$$= 75.705$$

So, 80A SP MCCB is needed from SDB to MDB

To Emergency Sub Distribution Board(ESDB):

In the garage,

$$I = (2*100 + 15*20)/(220*0.7)$$

$$= 3.24$$

So, 2*2.5 rm BYM + 2.5 BYA ECC are used

For lift,

$$I = (1*6000)/(220*0.7)$$

$$= 38.96A$$

So, 6*10 rm BYM + 2.5 BYA ECC are used

For pump,

$$I = (1*5000)/(220*0.7)$$

$$= 32.46A$$

So, 4*10 rm BYM + 2.5 BYA ECC are used

$$I = 3.24 + 38.96 + 32.46$$

$$= 74.66A$$

So, 75A SP MCCB is needed from SDB to MDB

For rooftop,

$$I = (2*20)/(220*0.7)$$

$$= 0.26A$$

So, 2*1.5 rm BYM + 1.5 BYA ECC are used